MTH 331(A) Dr. James Ashe
Exam 2 November 11, 2012 **Name:**

Determine the radius of convergence of the following power series.

1. $\sum \frac{x^k}{2^k}$

Use the geometric series $\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}$, *for* $|x| < 1$ *to find a power series representations for the following functions, and determine the interval of convergence.*

2. $f(x) = \frac{10}{1-x^2}$.

3. $f(x) = \frac{2x^2}{1-x}$.

Eliminate the parameter in following equations to obtain a single equation in terms of x *and* y.

4. $x = 2t$, $y = 4t^2$.

Determine ${}^{dy}/{}_{dx}$ *in terms of* t *and evaluate it at the given value of* t.

5. $x = 2t$, $y = 4t^2$; $t = 1$.

Give a set of parametric equations that describes the curve.

6. A circle centered at $(0, 0)$ with radius 6, generated clockwise.

7. *Express the following* ***polar*** *coordinates in* ***Cartesian*** *coordinates.*

8. $\left(-4, \frac{2\pi}{3}\right)$

Express the following ***Cartesian*** *coordinates in* ***polar*** *coordinates in at least* ***two*** *different ways.*

9. $(-1, 1)$

Locate the following polar coordinates on the graph.

10. A:$\left(2, \frac{\pi}{4}\right)$, B:$\left(-2, \frac{\pi}{4}\right)$, C:$\left(2, \frac{\pi}{4} + \pi\right)$, D:$\left(2, \frac{\pi}{4} + 2\pi\right)$

D:/home/jim/JamesAshe/JCSU/Calc_3_MTH_331/slides_Calc3/images/polar_graph_

Figure 0.1: Use this graph to plot the polar coordinates.

Mark the points on the ***polar*** *graph that corresponds to the points shown on the* ***Cartesian*** *graph.*

11. Mark the following points: A, B, C, D.

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Figure 0.2: The Cartesian graph is on the left and the polar graph is on the right.

Find the slope of the line tangent to the following polar curves at the given point using calculus.

12. $r = 2\cos 2\theta$; $\left(2, \frac{\pi}{2}\right)$.

Find the area of the region using calculus.

13. The region enclosed by all the leaves of the rose $r = 2\sin 4\theta$.